

What is it good for? An evidence-based review of stretching in sport and movement

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Introduction

Although its exact origins have not been scientifically validated, stretching is one of mankind's oldest health- and sport-related movements. Archeological excavations exposed millenia-old seals depicting yoga positions which substantially elongate the body's soft tissue (Lardner, 2001). Furthermore, the ancient Greeks performed flexibility exercises as a vital part of their conditioning programs when preparing themselves for the massive loads of the prestigious Olympic Games. To date, stretching continues to represent one of the most applied but, at the same time, most controversial maneuvers used to warm-up for, or recover from, physical activity.

In its most basic form, stretching is a natural and instinctive movement. After having a nap or resting for a longer period of time, cats and dogs frequently perform whole-body stretches before taking their first steps (Figure 10.1). Similarly, most people yawn and extend their joints prior to getting out of bed in the morning. Typical objectives of stretching in sports are injury prevention, performance enhancement and the acceleration of recovery. This chapter will describe the basic physiology, proven effects and limitations of related techniques in this context.

Basic assumptions of stretching interventions

Stretching always aims to elongate the soft tissue of the human body. If an applied force is large

enough to increase the length of a structure, two types of deformation can be distinguished. Elastic deformation is temporary and the original length is recovered shortly after the stretching force has disappeared. As an example, a



Figure 10.1

Big cats stretch too: a tiger gets ready for action

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rubber band returns to its original shape when it is now longer strained. By contrast, plastic deformation describes the permanent length change of an object. Wet chewing gum can be elongated to a tremendous degree but does not recover its original shape. It thus deforms plastically.

The reaction of the connective tissue to an elongating force can best be illustrated by means of the stress-strain curve of a tendon. The initially observed length changes during a stretch do not lead to a significant increase in strain because the collagen fibers, which are slightly wavy at rest, straighten. In this so-called toe region, there is no “real” tissue stretch and hence, little force is needed to achieve a length change. However, in the subsequent elastic or linear phase, the collagen is no longer crimped and starts to stretch progressively. More force is needed to induce tendon lengthening, which is associated with substantial increases in strain. When finally entering the zone of plastic deformation, the tendon may undergo irreversible length changes, potentially leading to rupture. While the stress-strain curve of other collagenous connective tissues such as ligaments or the deep fascia is similar to the tendon (Wang et al., 2009), the shape

can vary substantially in the muscle depending on whether it is actively or passively lengthened.

Muscle stretching: a misnomer

In most situations, stretching interventions affect both muscular and connective tissues. However, the most basic question practitioners and therapists need to ask themselves is which tissue is intended to be effectively stretched and if this is actually possible with the selected exercise. Contrary to traditional assumptions, stretching in joint end ranges does not always predominantly affect the muscle because other structures may represent the major limiting factor. For instance, inversion of the lower ankle joint stretches its lateral ligaments more than the peroneal muscles and elbow extension is limited by the bony structures of the joint rather than the biceps brachii muscle. Different types of stoppage are given in Table 10.1. In sum, stretching will always affect a variety of tissues and the main site of effect is dependent on the particular construction of the joint. The following section gives an overview about the differential effects of stretching on relevant soft tissue components of the locomotor system.

Table 10.1 Different types of stoppage	
Type of stoppage	Example (structure stopping movement)
Muscle	Hip flexion with extended knees (hamstring muscles) Ankle dorsal extension with extended knees (gastrocnemius muscle)
Bone	Elbow extension (olecranon of the ulna)
Soft tissue	Elbow flexion (biceps brachii muscle) Hip and knee flexion in obese people (posterior thigh/abdominal fat)
Ligament	Hip extension (iliofemoral ligament) Supination/inversion of the lower ankle joint (lateral ankle ligaments) Finger extension (diverse finger ligaments)
Capsule	Humeral rotation (capsule of the glenohumeral joint)

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Effects on the muscle

Looking at the micro-level, tissue elongation stretches the cytoskeleton of the sarcomere, the smallest functional unit of the muscle. Within the sarcomere, two contractile proteins, the thick myosin and the thinner actin, extend between its borders, the z disks. Under normal conditions, the myosin and actin filaments always overlap to a certain degree and the cross-bridges between both contribute not only to force production during contraction, but also to the resistance to stretch. Stretching can temporarily reduce the overlap of both filaments. However, an interesting mechanism ensures that sarcomere length is quickly restored: titin, a highly elastic protein connects myosin to the z disk. It acts like a spring, being unfolded upon tissue lengthening and pulling back the myosin filament into the resting state as soon as the stretching force has disappeared.

Immediately following stretching, changes in the muscle's mechanical properties can be observed. Static interventions with a total net duration of 2 to 8 min acutely reduce the stiffness of the gastrocnemius muscle (Ryan et al., 2008; Nakamura et al., 2011; Maeda et al., 2017; Bouvier et al., 2017) and the hamstrings (Hatano et al., 2017). By contrast, no such effect has been shown in the quadriceps muscle of the anterior thigh (Bouvier et al., 2017). Regarding the type of stretching, static lengthening seems to elicit larger decreases, despite dynamic stretching also being effective (Maeda et al., 2017). Between 10 to 20 min post-treatment, the stiffness values return to the baseline level (Ryan et al., 2008; Nakamura et al., 2011; Hatano et al., 2017). Interestingly, functional (enhanced range of motion, reduced peak torque) and subjective (increased stretch tolerance) alterations are maintained longer than the changes in mechanical tissue stiffness (Hatano et al., 2017). This observation

suggests that neuronal factors, such as a reduced cortical and/or spinal reflex excitability, contribute significantly to the effects of stretching.

The impact of chronic stretching exercise regimes on the muscle's mechanical properties has been investigated in a systematic review with meta-analysis. Including long-term interventions with durations of between 2 and 8 weeks (a minimum of two sessions per week), the authors concluded that stretching does not modify outcomes such as muscle architecture or stiffness (Freitas et al., 2018). It is hence noteworthy that the acute and chronic effects of stretching on the mechanical muscle properties are substantially different, although the high importance of sensory/neuronal adaptations (e.g. changed excitability and increased stretch tolerance) is a common feature of both applications.

Effects on the tendon

In addition to the muscle, the tendon can also undergo significant length changes. Upon passive stretching of the tibialis anterior muscle, Herbert and Gabriel (2002) found the contractile elements to account for only about 50% of the myotendinous unit (MTU) elongation. In the gastrocnemius, this share even decreased to 27%, suggesting that the tendon was lengthened considerably more than the muscle itself. These findings were confirmed by Morse et al. (2008), who reported a 53% proportion for the Achilles tendon. While both the tendon and the muscle are involved in MTU lengthening, the stretch duration seems to modify the ratio of their contribution. Kato et al. (2010) examined the effect of a static 20-min calf stretch on muscle and tendon length. Initially, both structures collectively caused the observed elongation of the myotendinous unit. However, only the tendon continued to lengthen after 10 min, whereas the muscle remained unchanged from then onwards.

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In contrast to the muscle, whose stiffness has been shown to decrease after or during a single bout of stretching, to date the immediate effects on the mechanical properties of the tendon are unclear. While Kubo et al. (2001) reported a 10% decrease following static stretching, Nakamura et al. (2011) detected an increase in the same outcome, and Kay and Blazeovich (2009) did not find any change at all. Possibly, the acute response is time-dependent, as Kubo et al., with 10 min, used the longest stretch duration compared with the other trials' 5 min (Nakamura et al., 2011) and 3 min (Kay and Blazeovich, 2009), respectively.

If chronic stretching interventions are considered, similar to the reaction of muscle tissue, there are no morphological changes in the tendon's mechanical properties following chronic stretching interventions of up to 2 months (Freitas et al., 2017). If structural effects are hence possible, exercises would have to be performed over a longer time frame.

Effects on the fascia

To date, the effect of stretching on the connective tissue surrounding the skeletal muscle has not been examined under human *in vivo* conditions. Notwithstanding, findings from fundamental research point towards a mechanical response of the fascia. When subjected to repeated isometric stretches, cadaver samples of the lumbar fascia initially reduce their stiffness. After a while, the values return to the baseline level and subsequently even increase above the pre-stretch state (Yahia et al., 1993). It was initially speculated that the activity of contractile cells may have caused this effect. In fact, fibroblasts and/or myofibroblasts can be found in the crural, plantar and lumbar fascia (Staubesand and Li, 1997; Schleip et al., 2019; Bhattacharya et al., 2010). However, two reasons suggest that factors other

than contractility represent the driving factor in acute, stretch-induced stiffness increases. The myofibroblasts located in the rat lumbar fascia reach their maximal contractile force only about 60 min following stimulation with mepyramine, a first-generation antihistamine (Hoppe et al., 2014). Furthermore, other animal experiments demonstrate that the strain hardening response of the tissue also occurs if the contractile cells are rendered unviable prior to the stretching intervention (Schleip et al., 2012). Contractility is therefore unlikely to represent the dominant mechanism of the rapid stiffness change. Instead, alterations of the water content seem to be a more important moderator: upon stretch, water is removed from the tissue and subsequently, similar to a sponge taking up new fluid, hydration increases during the rest period (Schleip et al., 2012).

Stretching and injury prevention: A fairy tale?

To stretch or not to stretch? This question is often discussed almost philosophically when it comes to the optimal preparation for physical activity. Frequently, flexibility exercise is implemented into warm-up and general prevention programs aiming to reduce injury risk. Notwithstanding, in contrast to wide-held beliefs, current evidence does not appear to support its application for this purpose. According to a series of systematic reviews, acute stretching interventions cannot decrease the overall occurrence of musculoskeletal traumas (Hart 2005; Small et al., 2008; Thacker et al., 2004). Herbert and Gabriel (2002) found only an almost trivial beneficial effect, pointing out that "the average subject would need to stretch for 23 years to prevent one injury". While these findings are discouraging at first glance, they may conceal an interesting effect of

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pre-exercise stretching. To date, most published research has focused on the absolute number of injuries sustained during sports participation. However, even opponents of stretching will agree that stretching cannot prevent unrelated disorders such as bone fractures. By contrast, it would be plausible to assume that athletes with a more compliant soft tissue following stretching are less likely to sustain a muscle strain injury (for the role of the connective tissue in this disorder, see Chapter 16). To adequately judge the helpfulness of stretching programs, it is, hence, imperative to focus exclusively on those health problems which may be affected by the performed intervention (e.g. muscle strain injuries only). Corresponding literature analyses have yielded contrasting results. For instance, Shrier (1999) reported no effect of stretching on the risk of muscle injury, but Small et al. (2008) suggested preliminary evidence for the effectiveness in the prevention of muscle-tendon disorders. McHugh and Cosgrave (2010) also examined the question as to whether or not stretching decreases muscle injury risk, including seven trials in their review. While the three studies (one of them in runners, where strain injuries would not be expected to be dominant) providing negative results registered only very few or no muscle strains, prevalence was higher in the four studies revealing a protective effect of the intervention. If a prevalence study does not register a high number of injuries from a specific type, it cannot be evaluated if an intervention would be effective in preventing them. It may thus be inferred that the studies not demonstrating a preventive value of stretching had only limited power to detect such effect. Finally, Behm et al. (2016) examined the findings of seven trials, which mainly included athletes performing running and sprinting activities. Pooling the results of the included studies, they found a risk reduction of 54% if stretching was

performed. The prophylactic effect of the intervention seemed to be particularly pronounced in the trials using longer stretch durations of 5 min or more. In summary, it can be concluded that stretching does not reduce the overall injury risk but may slightly reduce the incidence of muscle injuries. Determining the (time) cost-effect relationship of stretching for an individual is, hence, the most important task for therapists and trainers.

Although more a sensory correlate of training-induced adaptation than a musculoskeletal injury, delayed onset muscle soreness (DOMS) is frequently intended to be prevented by the use of stretching exercises. According to a systematic review of 12 studies enrolling a total of 2377 participants, stretching, on average, leads to an insignificant soreness reduction of half a point (pre-exercise) or 1 point (post-exercise) on a 100-point scale (Herbert et al., 2011). It can therefore be concluded that it does not represent an appropriate method to clinically decrease DOMS.

Stretch to win? Exercise effects on sports performance

Besides aiming to reduce injury risk, pre-exercise stretching is used to enhance performance. Depending on the chosen technique as well as the characteristics and goals of the respective sports, it can have both beneficial and detrimental effects for the athlete. If a large range of motion (ROM) represents a decisive criterion (e.g. in yoga, pilates or some elements of gymnastics), then the long- and short-term flexibility increases following stretching, which have been unequivocally documented (Behm et al., 2016), can be classified as performance-enhancing. Basically, all stretching methods (static, dynamic, antagonist-contract, contract-relax) are similarly effective, although the proprioceptive neuromuscular

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facilitation (PNF) methods appear to be slightly superior (Page, 2012). Regardless of the stretching type (static, dynamic or PNF stretching), the effects develop acutely and last about 30 min. The magnitude of the increases shows considerable variation (Table 10.2) and is dependent on several factors, such as the targeted joint and treatment duration. Stretch intensity (e.g. achieving no, mild or substantial discomfort/pain during the exercise) may also be a significant effect modifier; however, very few studies have explicitly tested or precisely reported this variable (e.g. Apostolopoulos et al., 2015). Data from two systematic reviews suggest that the combined application of stretching and heat (e.g. by means of ultrasound, laser or hot packs) are superior to a unimodal intervention (Nakano et al., 2012; Bleakley and Costello, 2013). By contrast, evidence is insufficient to recommend for or against the use of cold thermal agents such as ice (Bleakley and Costello, 2013).

Despite its beneficial impact on ROM, coaches and exercise professionals have to recall the potentially negative consequences of stretching in other performance-related outcomes. As soon as an athlete requires maximal or explosive

strength (e.g. powerlifters or sprinters), static stretching can have detrimental effects, causing performance decreases of up to 2% for sprints, 8% for jumps and 28% for absolute voluntary strength (McHugh and Cosgrave, 2010). Although compelling evidence clearly underpins the counter-productivity, many athletes used to performing stretches for many years do not want to break old habits. This is understandable when considering that it may also be of psychological value in the context of pre-exercise preparation. In these cases, there are different ways to modify but not forbid stretching. If the duration of a static stretch is reduced (particularly to less than 30 s), the decrements in performance shrink while ROM can still be enhanced, although to a smaller degree. A similar effect can be observed if stretch intensity is reduced (Behm and Chaouachi, 2011). In some cases, athletes will need both a large ROM and a high rate of force development. Consider a sprinter whose speed is dependent on explosive strength to create propulsion and a long stride to cover distance, which requires adequate hamstring flexibility. Reducing stretch duration and intensity may decrease the force loss but also the potential gains in stride length. In

Table 10.2 Mean range of motion improvements following stretching. The table lists data from systematic reviews					
	Static stretching	Dynamic stretching	PNF stretching	Diverse methods	Time frame
Radford et al., 2006	2.1 to 3.0° (ankle)				acute/ chronic
Harvey et al., 2002				8° (diverse joints)	chronic
Medeiros and Martini, 2018	5.2° (ankle)	3.8° (ankle, n.s.)	4.3° (ankle)		chronic
Decoster et al., 2005	5.7 to 28° (hip/knee)		5.3 to 33.6° (hip/knee)		acute/ chronic
Medeiros et al., 2016	8.4 to 12° (hip/knee)				acute/ chronic
Thomas et al., 2018	20.9% (diverse joints)	11.7% (diverse joints)	15.0% (diverse joints)		chronic
n.s. = non-significant change					

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other words, if ROM is as important as explosive power, then the described modifications could thwart the potential benefits. As many sports do require both qualities, the solutions of reducing intensity and duration are not optimal. In addition to static stretching, extensive research has therefore been dedicated to its dynamic form, which does not substantially reduce and in some cases even slightly improves functional performance (McHugh and Cosgrave, 2010; Behm and Chaouachi, 2011). In sports involving quick powerful actions based on stretch shortening cycles, such as football (soccer), basketball or some athletic disciplines, this form of stretching seems preferable.

Out of balance? Stretching and muscular asymmetry

A frequent health-related goal of trainers treating elite and regular athletes is the reduction of muscular imbalance. The classic concept of asymmetry can be best compared to a sailing vessel. In order to properly function, its mast has to remain uprightly centered in perfect balance, thanks to the rigging ropes holding it from different sides. In our body, the mast would correspond to the bony structures, i.e. the spine, while the ropes represent the muscles and other soft tissues. If no imbalance is present, the muscles help to ensure the maintenance of the typical and optimal curvatures (lumbar lordosis, thoracic kyphosis and cervical lordosis). However, if one component, the pectoral muscles or the forestay (the cable which pulls the mast in the anterior direction) are too strong relative to the opposite side (our upper back muscles or the cable pulling the mast backwards), the spine/mast will bend in their direction. Over time, this can lead to overuse and structural damage. Besides the upper spine, where the dominance of the pectoral muscles leads to hyper-kyphosis,

another classical example of muscle imbalance relates to the lumbopelvic region. Here, weak abdominal/gluteus maximus muscles and overactive hip flexor/lumbar extensor muscles create an excessive lumbar lordosis. In some sports, lumbar lordosis is particularly frequent. Nine out of ten dancers and gymnasts exhibit moderate or strong lordosis, which is necessary for some recurrent movements (Ambegaonkar et al., 2014). However, if pain develops in the low back region, the use of corrective “de-lordosing” exercise should be considered.

The traditional treatment of muscular imbalance includes two pillars. First, the weakened muscles are strengthened/activated to counteract the bending forces. Second, the overactive/excessively strong muscles are stretched in order to decrease their tone, which should assist the strengthened antagonists in pulling the joint back in balance. However, as outlined in this chapter, a permanent stretch-induced reduction of resting tension is not possible due to the spring-like function of the titin-filaments, which restore original tone in the minutes following a stretch. Does this mean that stretching should no longer be performed if a person has bad posture caused by muscle imbalances? The answer is both yes and no. This is best illustrated by an example. If an athlete has protracted shoulders and a kyphotic thoracic spine, a basketball player, for example, they will not be able to fully extend the latter. If going up for a rebound or shot, they have to compensate in the neighboring joints (the lumbar and cervical spine, possibly also in the shoulder joint), which may cause problems and performance deficits. In the long term and under static conditions, stretching will not help because the effect of the prescribed pectoralis stretching always diminishes after a while. The preferred chronic correction strategy is, therefore, rather the strengthening of the weakened

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antagonists (upper back muscles). However, in the short term, stretching can still make a lot of sense: after acute bouts of stretching the pectorals, the athlete may benefit from their temporarily reduced tone to play their stint on the field without excessive protraction before being substituted. Off court, they could use the briefly corrected posture to train without and into less compensatory movement patterns.

Stretching in the rehabilitation of musculoskeletal disorders in sports

Despite its widespread use in the rehabilitation of sports-related musculoskeletal disorders, few studies have explicitly examined the value of therapeutic stretching, and in most studies, related techniques were used solely as an additional treatment. The following overview should therefore be interpreted with caution. According to one trial, stretching exercises can increase the treatment success of eccentric training in patellar tendinopathy, which develops particularly in athletes who participate in jumping and running sports. Dimitrios et al. (2012) randomly allocated 43 patients to two groups exercising five times per week. While the individuals performing eccentric for the quadriceps femoris only increased the scores of the VISA-P questionnaire (higher scores reflect lower pain levels) from 46 at baseline to 74 after 4 weeks and 77 after 6 months, additional static stretching of the anterior thigh was shown to be significantly more effective (44 to 86 to 94 points). Similarly, stretching can be used as an adjunct therapeutic element in myofascial pain syndrome. Being characterized by the presence of active myofascial trigger points, the pathology often affects overhead athletes and manifests as neck pain. Following prior deactivation of the trigger points, for example, by means of acupuncture, it has been found that stretching further increases the analgesic treatment effect

when compared with a placebo intervention (Wilke et al., 2014). In contrast to these studies, compelling evidence is available for effectiveness in treating posterior shoulder tightness (ST) and glenohumeral internal rotation deficit (GIRD), both of which may predispose for upper extremity injury in sports or chronic impingement syndrome (Harshbarger et al., 2013; Johnson et al., 2018). Athletes typically affected by ST and GIRD include throwers (e.g. in baseball), swimmers or tennis players. According to two systematic reviews, a cross-body arm adduction stretch has a positive impact and can be recommended (Harshbarger et al., 2013; Mine et al., 2017). By contrast, the often applied sleeper stretch does not necessarily improve ST and GIRD (Mine et al., 2017).

How to do it: the different stretching methods

Over the years, different types of stretching have evolved (see Table 10.3). The most traditional methods include static and dynamic stretching. In static stretching, a constant joint position, which elongates the targeted soft tissue, is maintained without any further movement. Dynamic stretching describes repetitive movements providing cyclic lengthening stimuli, most frequently at the end range of motion (then called ballistic stretching).

In addition to static and passive stretching, newer methods aim to specially engage the central nervous system using PNF (Figure 10.2). A common feature of PNF stretches is that the deliberate stimulation of sensory receptors is suggested to trigger muscle inhibitions, which then allow a higher increase in flexibility.

- **Antagonist contraction (AC) stretching:** To actively flex the knee, the hamstring muscles need to be contracted. Simultaneously,

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for a fluent and energy-efficient movement, it is necessary to relax the quadriceps femoris muscle, because major activity of this antagonist would decelerate or stop the flexion motion. Reciprocal inhibition represents the mechanism helping to put this principle into practice: if the hamstrings are contracted, this information is sent back to the central nervous system, which then inhibits the drive to the quadriceps muscle. AC stretching is based on reciprocal inhibition. When stretching the ventral thigh (and with this the quadriceps femoris), the hamstrings are simultaneously contracted with the aim to trigger the process described above.

- **Contract relax (CR) stretching:** As one of the most important proprioceptors, the

Golgi tendon-like organ continuously registers the contraction state of a skeletal muscle. When the quadriceps femoris is activated, the receptor fires, activating an inhibitory neuron in the spinal cord. As a consequence, the contraction of the quadriceps is released. This process is called autogenic inhibition. In CR stretching, the target muscle (in our example again the quadriceps femoris) is strongly contracted for a few seconds. During the subsequent stretch, the quadriceps can be lengthened to a stronger degree thanks to autogenic inhibition.

- **AC-CR stretching** combines the two PNF methods described above. Again, the quadriceps femoris serves as an example. Initially, the quadriceps is contracted to

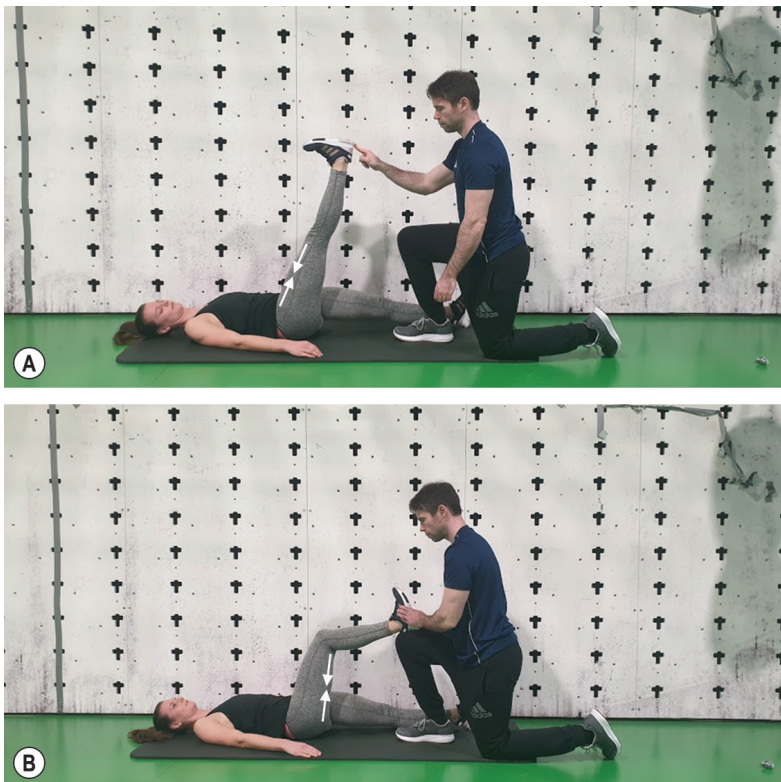


Figure 10.2

Besides static and dynamic stretching, proprioceptive neuromuscular facilitation (PNF) methods can be used by more experienced clients. In (A) antagonist contraction (AC) stretching, the antagonist (white arrows) is contracted during the exercise. For (B) contract relax (CR) stretching, the target muscle is isometrically activated (white arrows) immediately before the actual stretch. AC-CR stretching combines both methods.

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Table 10.3 Technique-effect matrix for different stretching methods. The table shows synthesized results from systematic reviews

	Static		Dynamic		PNF	
	acute	chronic	acute	chronic	acute	chronic
Range of motion	↑	↑	↑	↑	↑	↑
Maximum strength	↓	↑	↔/↑	↔/↑	↓	↔
Explosive strength (jump height/sprint)	↓	↔/↑	↔	↔	↔	↔
Injury risk	↔/(↓)	↔/(↓)	↔/(↓)	↔/(↓)	↔/(↓)	↔/(↓)

Key: ↑ = increase, ↓ = decrease, ↔ = no change or unknown

elicit an autogenic inhibition releasing its tension. Immediately following this step, a stretch with simultaneous contraction of the antagonist (the hamstrings) is performed, which causes a reciprocal inhibition of the quadriceps. In summary, AC-CR stretching therefore uses two PNF pathways to more strongly decrease the activation of the target muscle.

The pros and cons of using PNF stretching in clients and patients need to be carefully weighed against each other. Due to their complexity, the techniques are more difficult to apply in some populations (particularly in elderly persons or individuals with limited sport-related experience) and can require more time. However, despite these drawbacks, they may also offer variety, refreshing an old training routine or helping an elite-level athlete to increase performance by a few but decisive percent (Table 10.3).

Summary

Stretching is a vital component of general fitness programs as well as strength and conditioning routines. However, its value is highly ambivalent and context-specific: while stretching substantially improves joint range of motion regardless of the technique used, it does not, or only to a minor degree, prevents sports injuries and, furthermore, cannot reduce muscle imbalance permanently. When warming up for explosive and powerful movements, athletes should use dynamic over static exercises as the latter may have detrimental effects. Despite extensive efforts, the mechanisms behind stretching have not been fully elucidated. Acutely, changes of the mechanical properties in the muscle, tendon and fascia become manifest. However, they may not be preserved in the long term. Besides morphological adaptations of the stretched tissues, the role of the central nervous system (e.g. by reducing stretch tolerance) continues to play an important role in the genesis of the effects.

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